SHANIYA JARRETT

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SUMMARY

Former engineer and current PhD researcher at NASA Goddard **seeking a data science, machine learning, or product management internship**. 7+ years of technical experience, specializing in deep learning applications. Track record of executing end-to-end projects, from delivering a $65M hardware system to turning a terabyte-scale analysis into published findings.

EDUCATION

University of Maryland — *Ph.D., Astrophysics & Astronomy* 2024 - Present
Fisk University — *M.S., Physics* 2022 - 2024
University of Colorado, Boulder — *B.A., Astrophysics* 2014 - 2020

SKILLS

Hard Skills: Python, SQL, SciPy, Scikit-learn, Pandas, PyTorch, Hugging Face, Transformers, CNNs, Git/GitHub, Atlassian Tools, SolidWorks, Zemax, LaTeX, AWS, APIs, Tableau.

Soft Skills: Cross-team Communication, Collaboration, Open-ended Problem Solving, Adaptability, Initiative, Creativity, Strategic Thinking, Project Management, Receptive to Feedback.

Certifications: AI Agents in Copilot, Microsoft (2026) | Agile Project & Risk Management, NASA, IACET-accredited (2026)

EXPERIENCE

Scientific Researcher **2022 - Present**

NASA Goddard Space Flight Center / University of Maryland / Vanderbilt University

- Built a signal classification transformer (LLM architecture) using millions of time-series datasets to find high-priority targets for a **$1.5B NASA mission**, cutting false positives **3x** compared to standard methods, reducing operating costs.
- Fine-tuned a pre-trained audio/speech recognition transformer from OpenAI using GPU acceleration to develop an ML pipeline for low signal-to-noise data, enabling transparent visualization and insight into model decisions.
- Diagnosed and parallelized a bottleneck in the training data pipeline, resolving memory failures and **speeding up training 6x**; benchmarked memory scaling to size supercomputer runs.
- Validated model results with frequentist and Bayesian statistics, including hypothesis tests and likelihood-based model comparison, and ran ablation, saliency, and spurious-correlation checks to rule out artifacts.
- **Took a project from concept to publication**: applied correlation and clustering analysis to **1 TB** of simulation data, discovering that a rare event could serve as a low-cost proxy for predicting trends across a much larger dataset.

Laboratory for Atmospheric and Space Physics (LASP) ***2017 - 2020***

- Developed and adapted anomaly detection and trend analysis algorithms for 4 NASA missions using statistical and signal-processing techniques, discovering a nonlinear relationship between two seemingly unrelated features and using it to build a predictive model of a key target variable.
- Optimized satellite imaging with ray-tracing simulations, reducing light scatter and **sharpening images by 20%.**

Satellite Test Engineer **2020 - 2022**

General Atomics, Centennial, CO

- Planned and led the vibration, mass-properties, and structural test campaign for the **$65M Gazelle spacecraft**, collecting and validating performance data against requirements to meet every delivery milestone.
- Tested 5+ subsystems across hardware and software; co-pioneered the company's first thermal-vacuum campaign, establishing qualification standards later adopted across future missions.
- Partnered with program management to resolve schedule delays by reprioritizing tests and reallocating resources.
- Translated test results and insights into actionable recommendations and **presented results to cross-functional engineering teams and stakeholders,** informing design fixes before launch.

OUTREACH PROJECTS

Program Manager / Creator - AstroBeats Project (astrobeats.us) **2023 - 2025**

- Designed a 4-part data-driven workshop turning real astrophysics data into music: wrote the proposal, won competitive grant funding, built the scope, schedule, and budget, and scaled it into a full program, earning a university leadership award, two national conference talks, and recognition from Vanderbilt, UMD, and multiple podcasts.

Instructor & Workshop Guide **2023 - 2024**

- **Foundations of Astronomical Data Science:** taught participants to query large research databases with SQL and ADQL.
- **Black Girls Becoming (BGB) Summer Institute:** developed and taught a science curriculum alongside Vanderbilt faculty for a cohort selected from 55 applicants, contributing to a 9-year study on STEM outcomes for participants.

SELECT AWARDS & FELLOWSHIPS

Outstanding Teaching Assistant Award (2025) • Margaret Cuninggim Women's Center Leadership Award (2024) • CMNS Exceptional Recruitment Award (2024) • NSF Graduate Research Fellow (2024) • Heising-Simons Mini-Grant Winner (2023)